

Introduction

DS205.3 – Data Science in Python

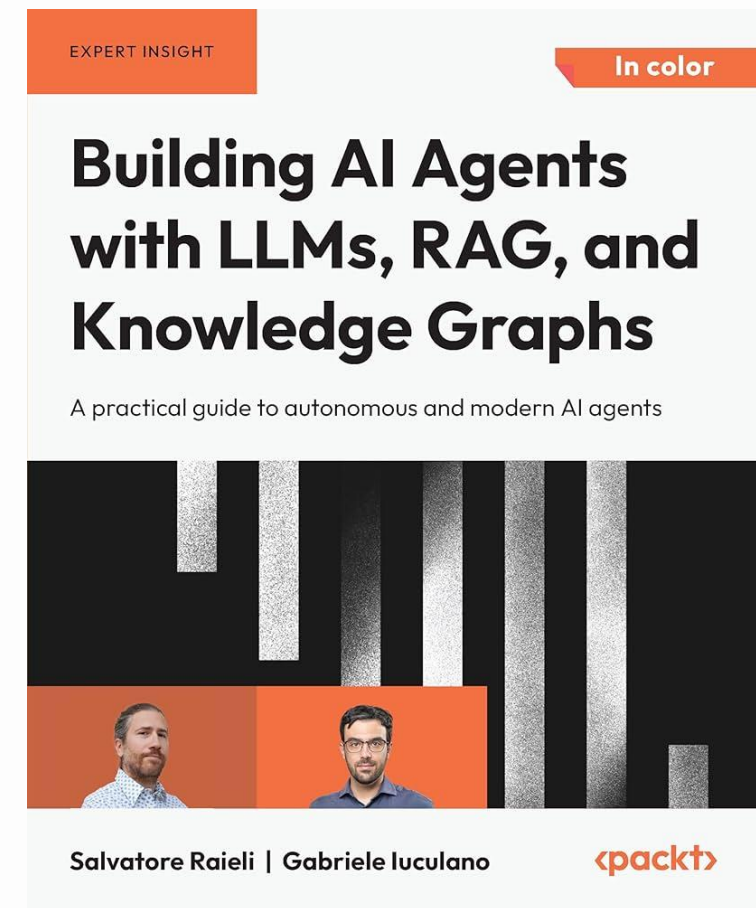
Mr. Anton Jayakody

MSc (CS)[R], BSc (CS)



Recommended Reading

- Lawson, B. (2024) Building AI Applications with Python: LLMs, RAG, and Agents. Sebastopol: O'Reilly Media.
- McKinney, W. (2022) Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter. 3rd edn. Sebastopol: O'Reilly Media.



Content

- Data Pipeline: A series of automated processes that move data from a source to a destination, undergoing validation and transformation along the way.
- Asynchronous Programming: A coding paradigm that allows a program to handle multiple tasks concurrently. Essential for high-performance API/Data integration.
- Type Hinting: A formal way to indicate the data type of variables and function parameters. This enables IDEs to provide autocomplete and catch errors during development.
- Defensive Programming: A design approach where code anticipates potential errors and handles them gracefully without crashing the entire system.

Type-Hinted Function

```
def fetch_data(source_url: str) → dict:  
    # We promise to return a dictionary  
    return {"status": "success", "data": [1, 2, 3]}
```



Asynchronous Execution

```
import asyncio
```

```
async def fetch_source1(name: str) → str:
    await asyncio.sleep(1)
    # Simulates network lag
    return f"Data from {name}"
```

```
async def fetch_source2(name: str) → str:
    return f"Data from {name}"
```

```
async def main():
    # Run multiple fetches at once
    results = await asyncio.gather(
        fetch_source1("Database"),
        fetch_source2("API_Endpoint")
    )
    print(results)
```

Result: Data from API_Endpoint
Data from Database

Exercise

1. Define a function `Calculate_average(data: list[float]) → float`. If the list is empty, return 0.0. Ensure you include the type hints for both the parameter and the return value.
2. Create a function `simulate_api_call(user_id: int)` that uses `await asyncio.sleep(0.5)`. Call this function 3 times using `asyncio.gather` and measure how much time it takes to finish.
3. Write a script that takes a list of integers as strings. Use a try-except block to convert them to integers.